

Logic Gates — Paper D

HARD • NON-CALCULATOR • 45 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. The mark for each question is in square brackets on the right.
- No calculator is needed anywhere on this paper.
- Truth-table rows go in **binary counting order**: 00, 01, 10, 11, and for three inputs 000, 001, 010, 011, 100, 101, 110, 111.
- Write expressions with AND, OR, NOT, NAND and NOR, and put **brackets wherever two different gates meet**. NOT applies only to the thing straight after it, so NOT A AND B means (NOT A) AND B.
- Several questions describe machines you may not have seen before. You are not expected to recognise them: apply the gate rules and the one-gate-at-a-time method you already have.
- No reminder of the gates is printed on this paper.

SECTION 1 – QUICK AND EXACT

1. Complete the trace table for $Q = (A \text{ NOR } B) \text{ NOR } (\text{NOT } C)$. P is the output of the first gate and R is the output of the NOT. [3]

A	B	C	P A NOR B	R NOT C	Q P NOR R
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

2. Answer these without writing out the tables. Rows are numbered from the first row, so the first row of a three-input table is row 1 (000). [1]
- (a) In a three-input table (A, B, C), write the **sixth** row. [1]
- (b) In the same table, which row number is $ABC = 110$? [1]
- (c) In a four-input table (A, B, C, D), write the **tenth** row. [1]

3. A gate has two inputs, and its output column contains exactly **three** 1s.

(a) Name the two gates this could be.

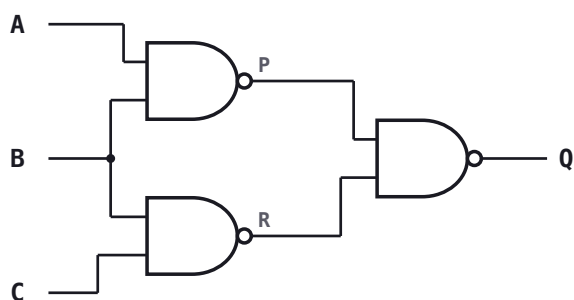
[2]

(b) Write down one input row (AB) that would tell you which of the two it is.

[1]

SECTION 2 – CIRCUITS AND TABLES

4. Study the circuit, which is built only from NAND gates. P and R are the outputs of the first two gates.



(a) Write the expression for Q.

[2]

(b) Complete the trace table.

[3]

A	B	C	P	R	Q
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

5. Complete the table for these three expressions, which are easy to confuse:

$$X = \text{NOT } (A \text{ AND } B) \quad Y = (\text{NOT } A) \text{ OR } (\text{NOT } B) \quad Z = (\text{NOT } A) \text{ AND } (\text{NOT } B)$$

(a) Complete the table.

[3]

A	B	X	Y	Z
0	0			
0	1			
1	0			
1	1			

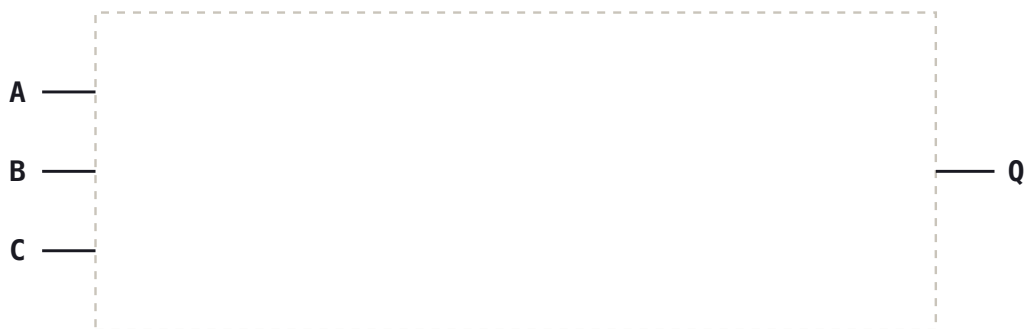
- (b) Two of the three columns are identical. Say which two, and name the single gate they equal. [1]

SECTION 3 – MACHINES AND DIAGNOSIS

6. At a school in Pune the period bell, Q, rings when the timer fires, **unless both** the exam-hall switch and the silent-mode switch are on. **A = 1** when the timer fires. **B = 1** when the exam-hall switch is on. **C = 1** when the silent-mode switch is on. If B and C are both on, the bell must stay silent. If only one of them is on, it must still ring.

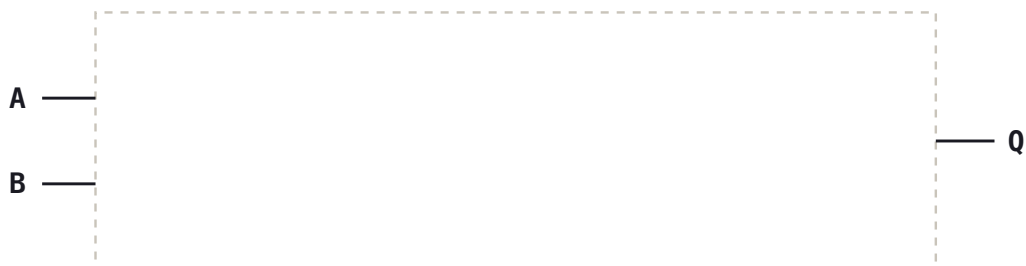
- (a) Write an expression for Q. [2]

- (b) Draw the circuit in the box, using the standard gate symbols. [2]



7. The store only has **NAND** gates. Draw an **AND** gate made from NAND gates only, then check it.

- (a) Draw the circuit in the box. Label the output of the first gate P. [2]



- (b) Complete the trace table for your circuit. [1]

A	B	P	Q
0	0		
0	1		
1	0		
1	1		

8. Rohan has written two statements in his notebook, and **both are wrong**. For each [5]
one, explain the mistake and write a correct version.

ROHAN'S NOTEBOOK

(i) "A NAND gate is an AND gate with a circle added, so it gives 1 when both inputs are 1."

(ii) "NOT A AND B means the same as NOT (A AND B), because the NOT comes first."

End of paper. Check every truth table has all its rows in counting order, and that every expression has brackets wherever two different gates meet.